Threaded Binary Tree

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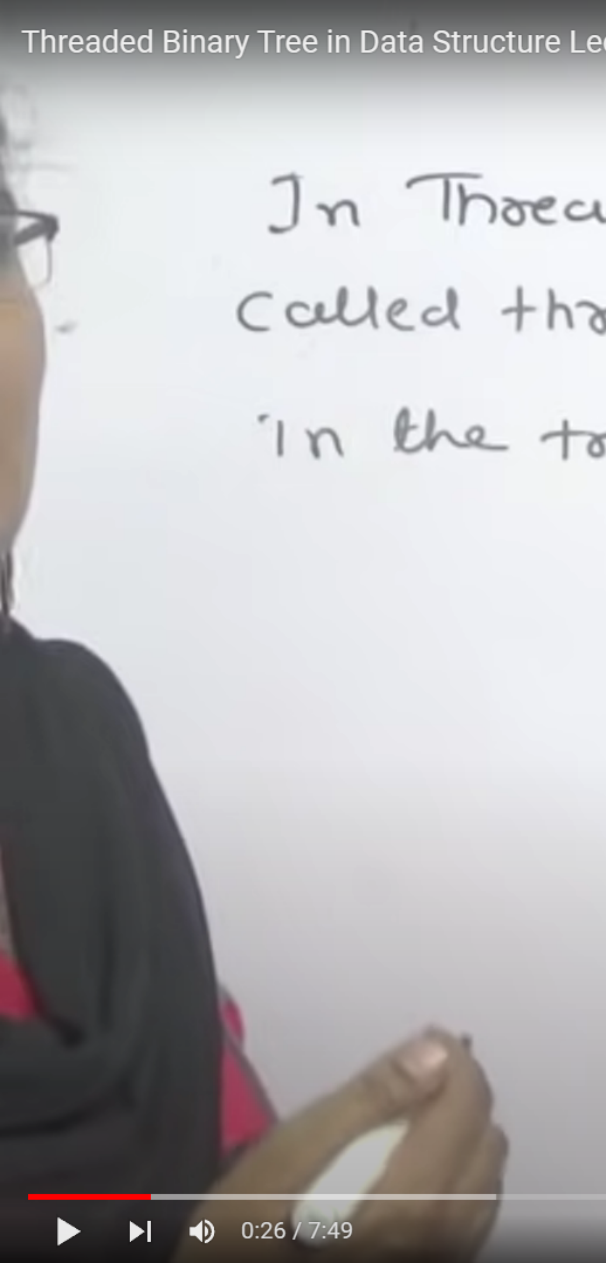

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Inorder

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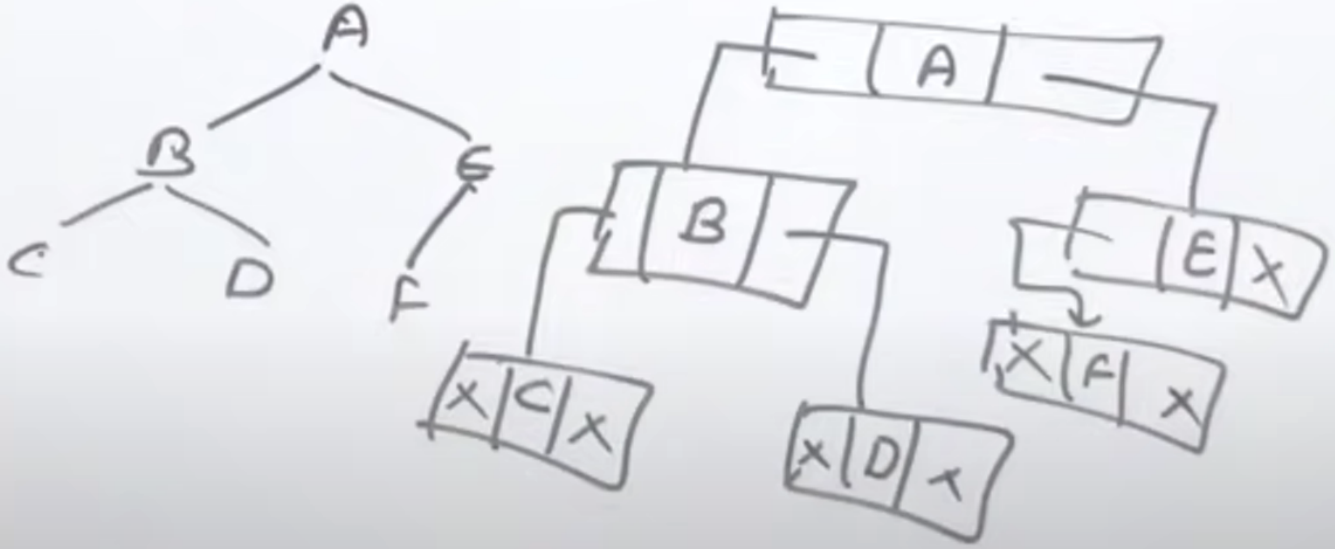
CBDAFE



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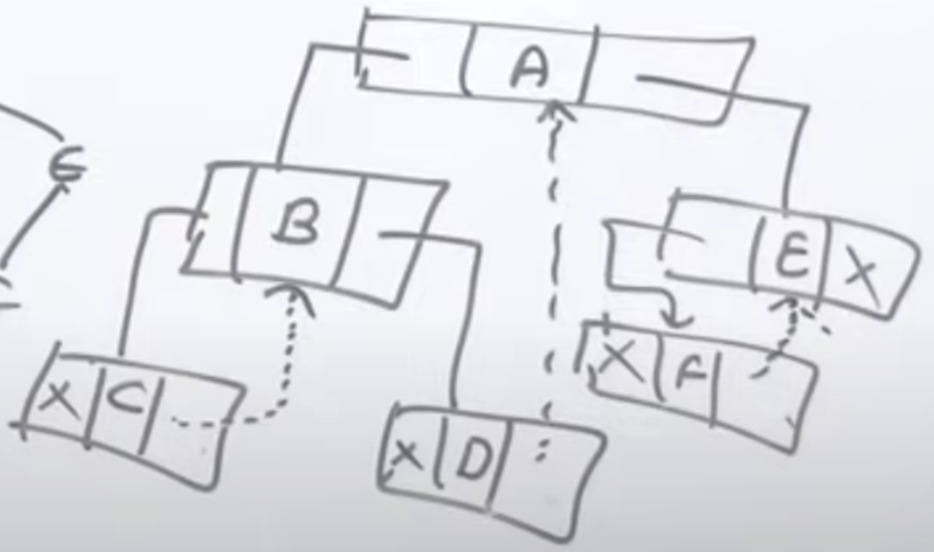

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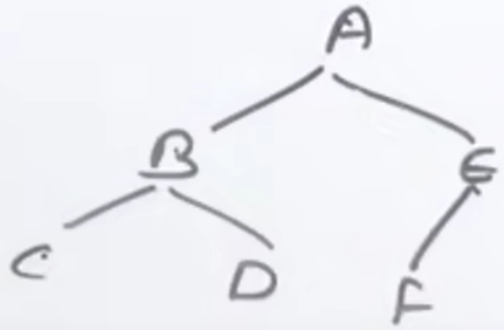

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1. One way Threading
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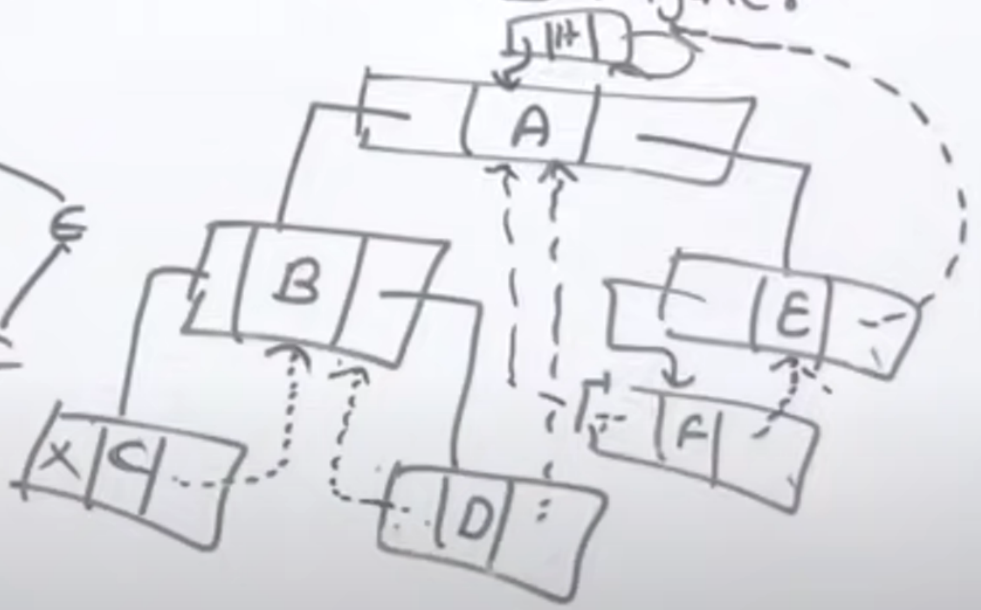


M way tree

* (M-1) key values per node
* M subtrees

where M= degree of tree.

Example:

BST

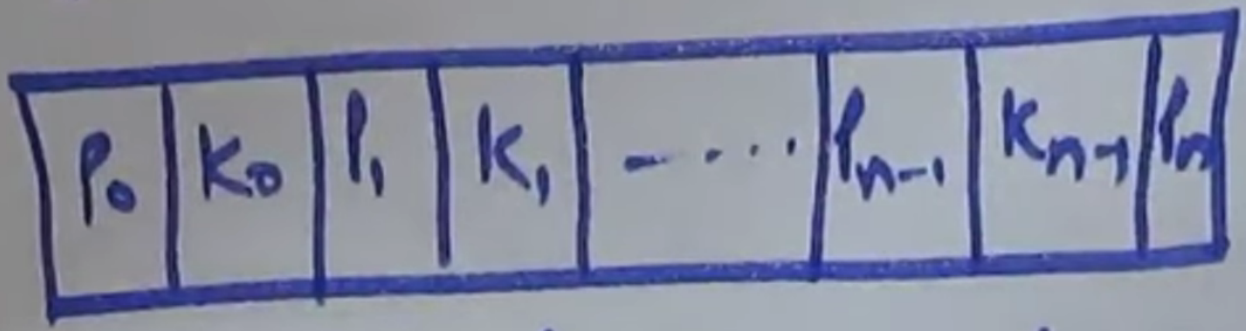
M= 2

(one value per node and

Properties of M-way Trees | Data Structres | M-way Search Trees (in HINDI)

= 2

'M' 'N'



(N-1) values

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$K_0 < K_1 < K_2 - - - < K_{n-1}$

4:10 / 7:13

M-Way Search Tree

↳ M is a fixed upper defines the no. of values that can be stored in a node.

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No. of key values

Properties of M-way Trees | Data Structres | M-way Search Trees (in HINDI)

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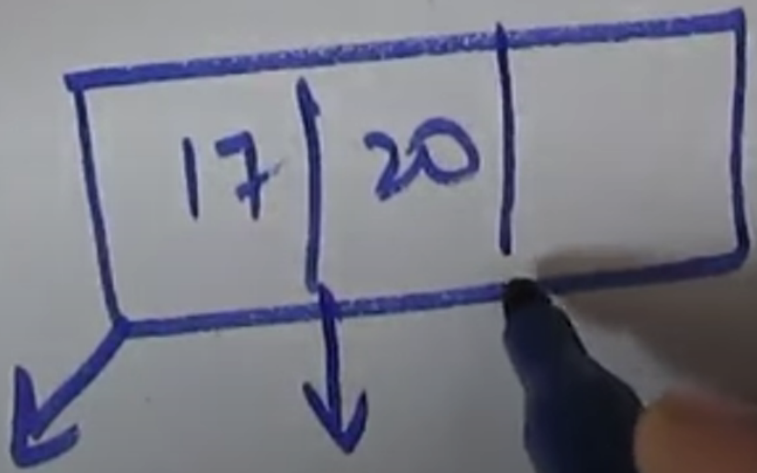
i → values ⇒ (i+1) subtrees

M=5

4 values



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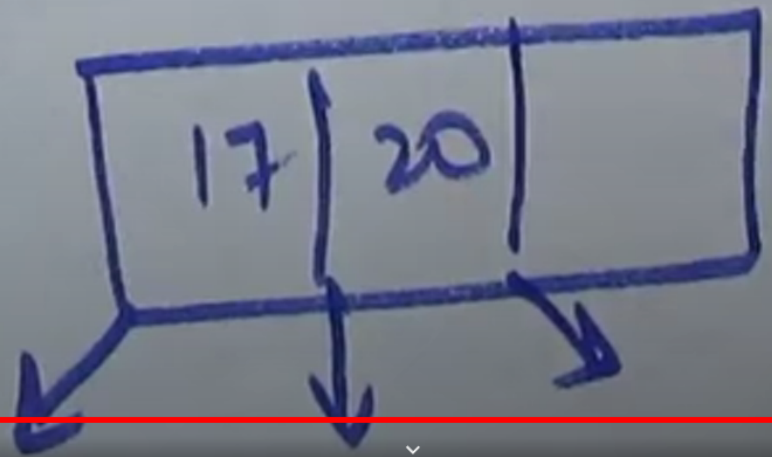
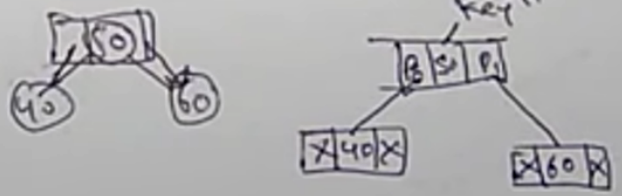


6:39 / 7:13

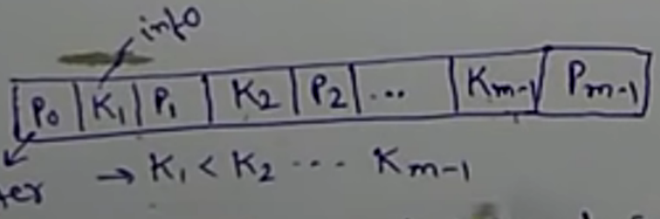
(Multiway) M-Way Search Tree

⇒ It is similar to Binary search Tree but it has m pointers (child) and m-1 key in one node (x)



M-Way Search Tree properties:

1) Each node can point at most m subtree
2) The number of keys in each node are maximum m-1 and each node can have maximum m pointer
3) m-way tree representation



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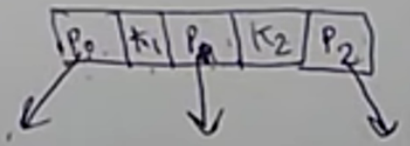
3-way →



$K_1 < K_2$

m subtree

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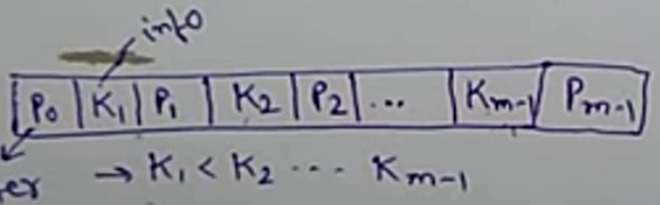

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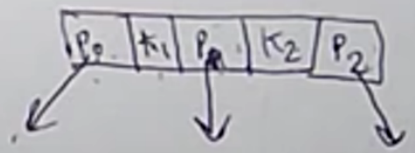

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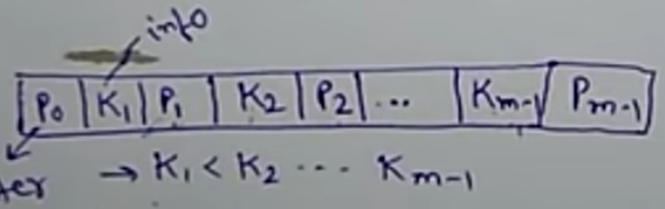

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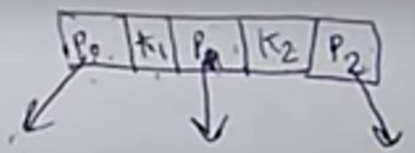

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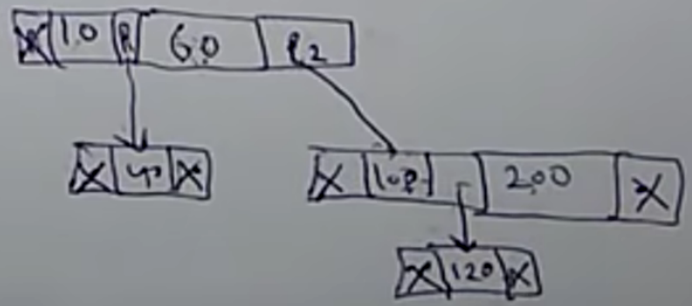
Ex 10, 60, 100, 200, 40, 120

$m=3$



B-tree

- balanced m-way tree (m: order)
- Generalization of BST in which a node can have more than one key & more than 2 children
- maintains sorted data
- all leaf node must be at same level
- B-tree of order m has following properties:
  - → Every node has max. m children
  - → Min. children :- leaf → 0
    root → 2
    internal nodes → $\lceil \frac{m}{2} \rceil$
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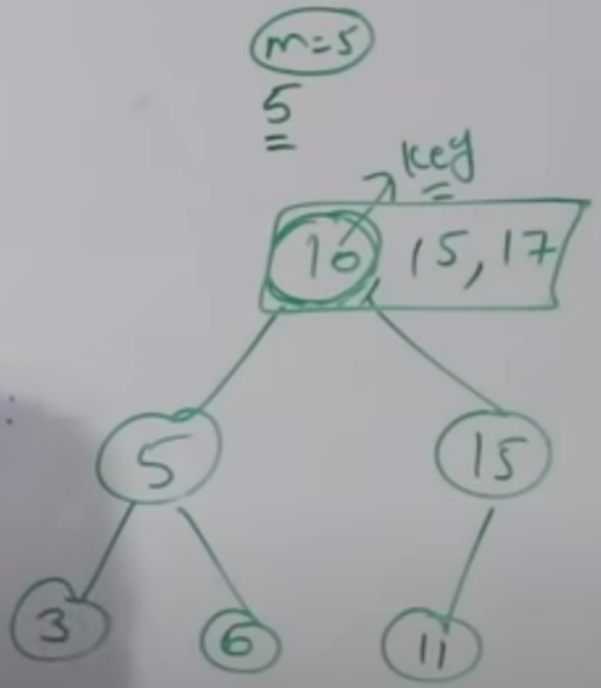

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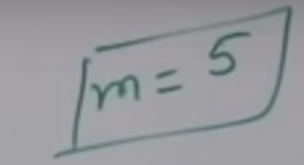

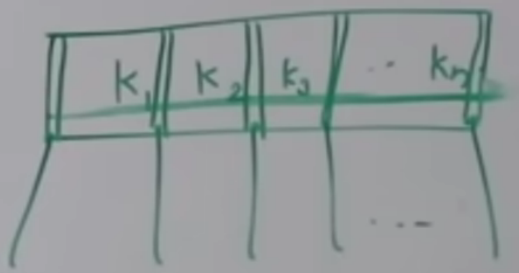

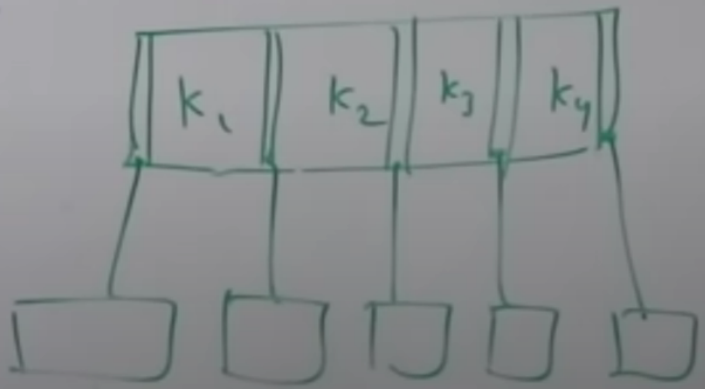

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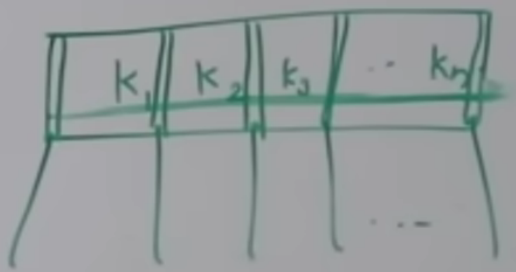
m = 5



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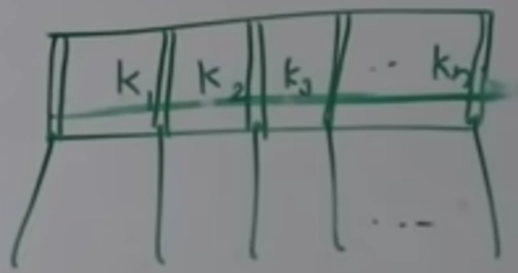

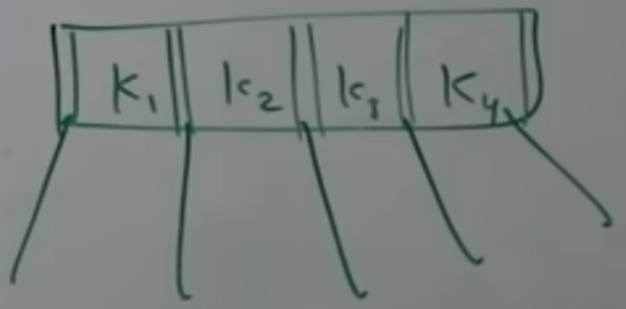

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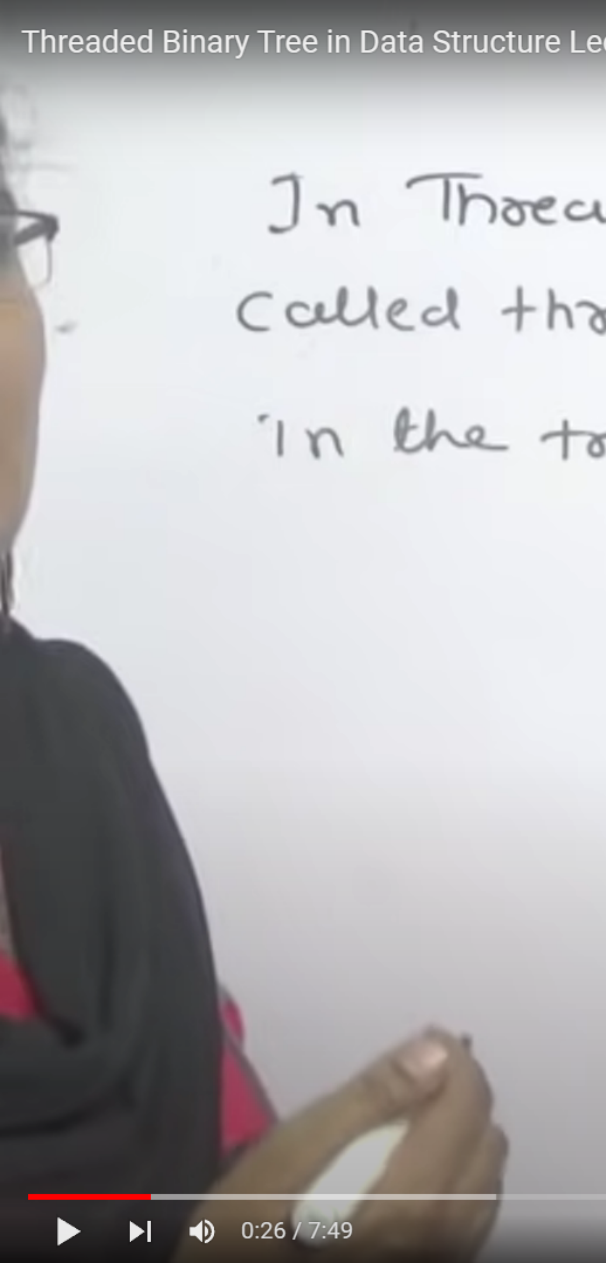

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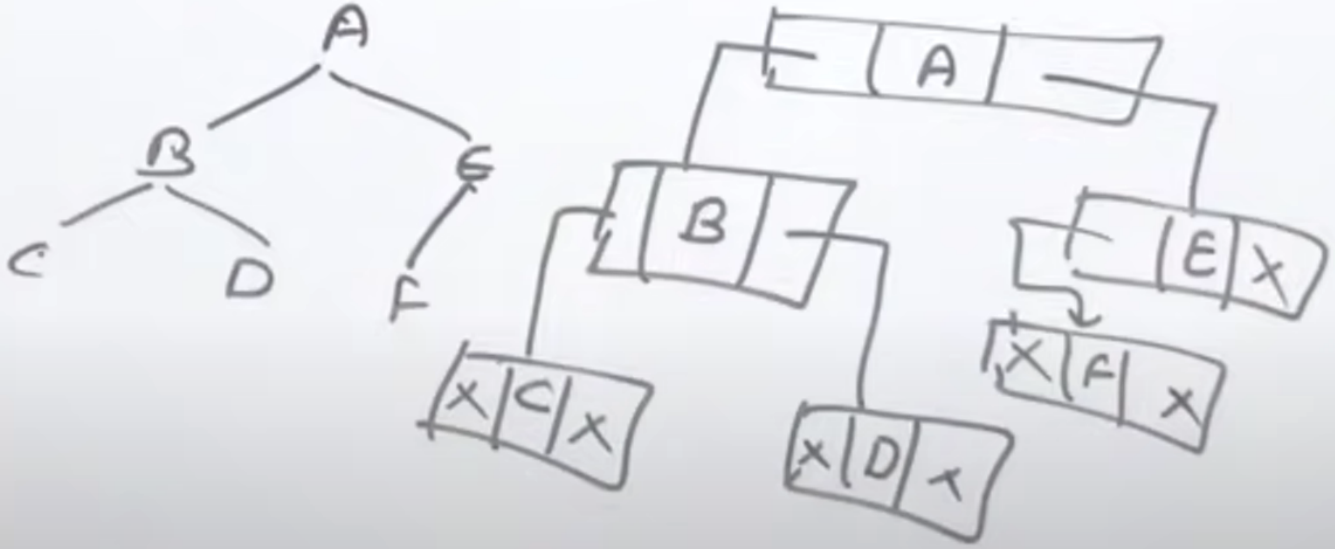
CBDAFE



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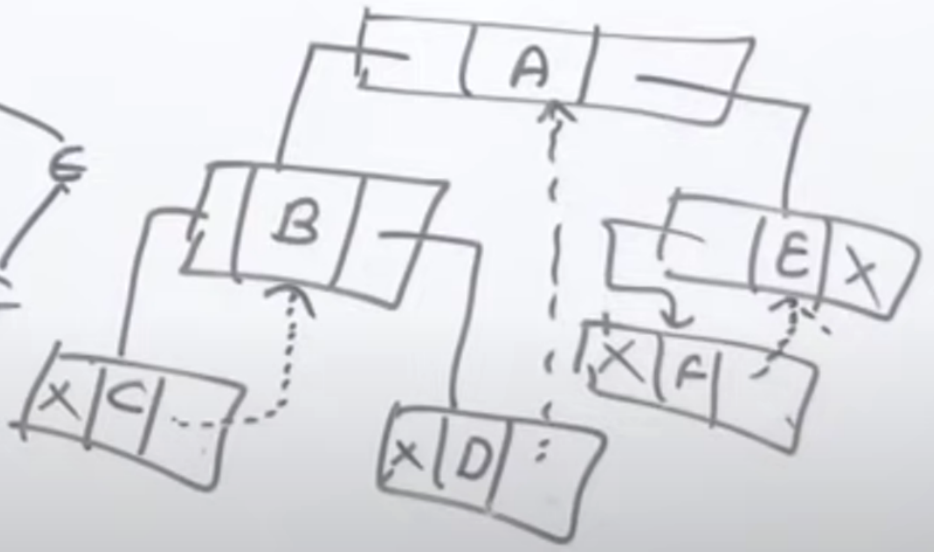

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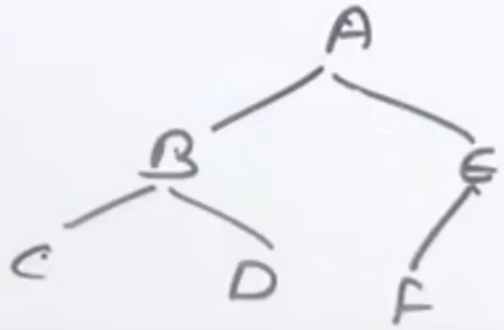

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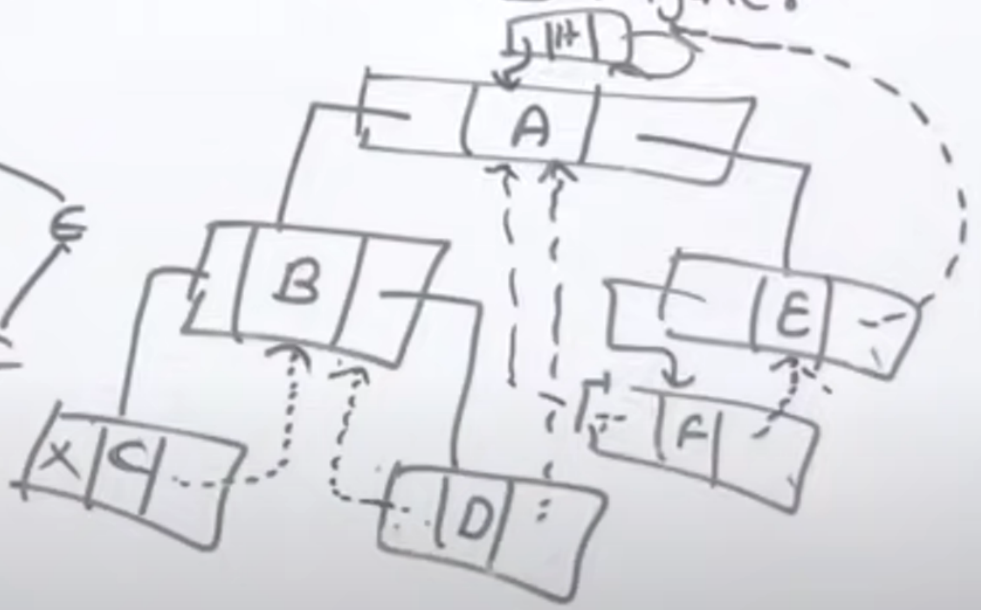

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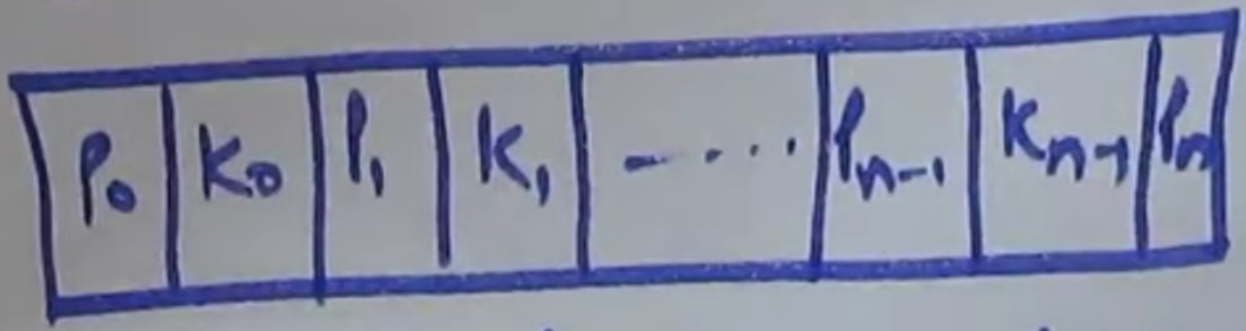

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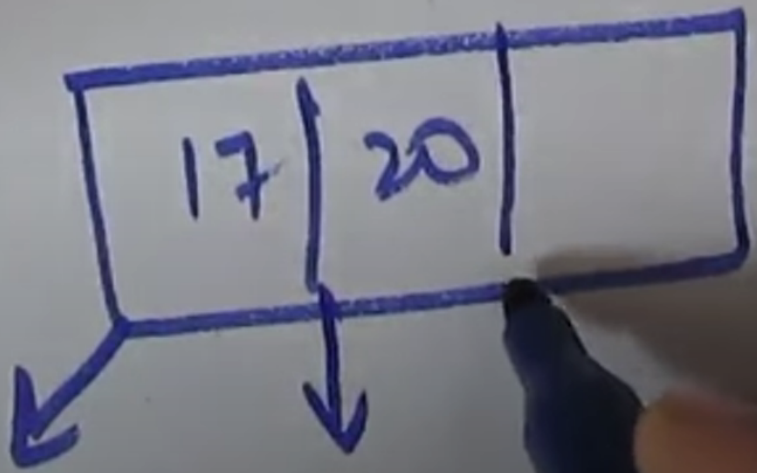

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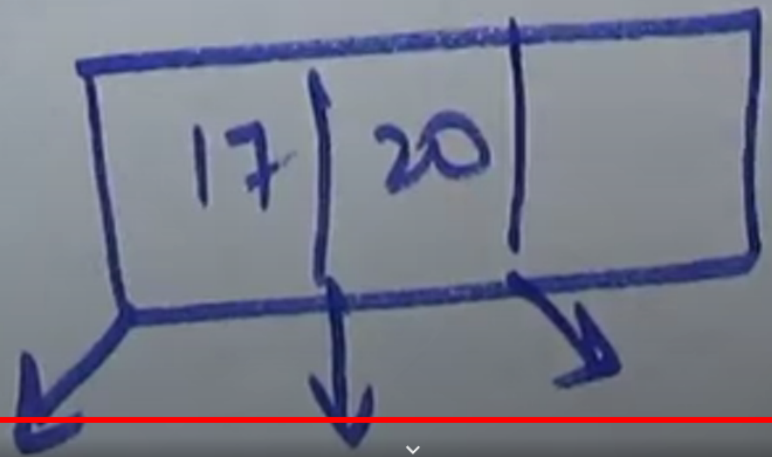

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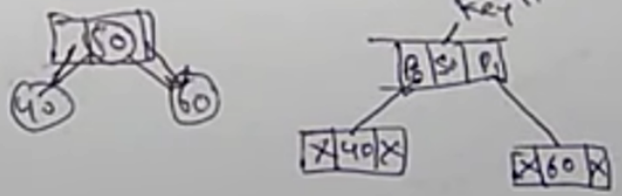

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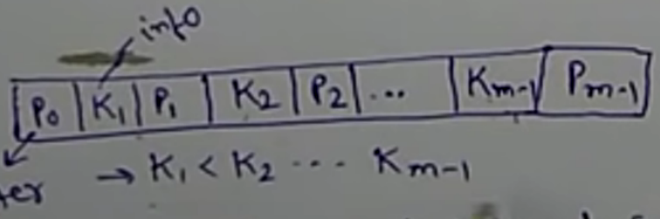

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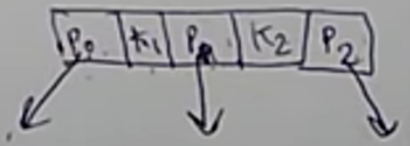

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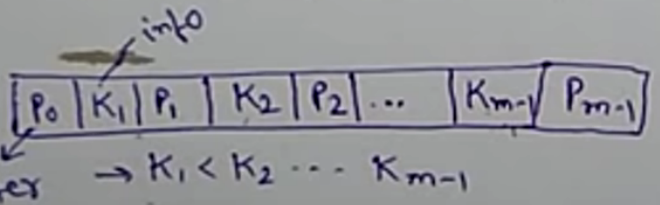

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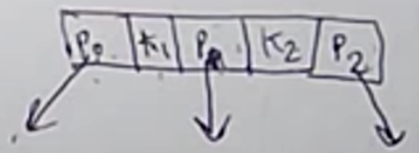

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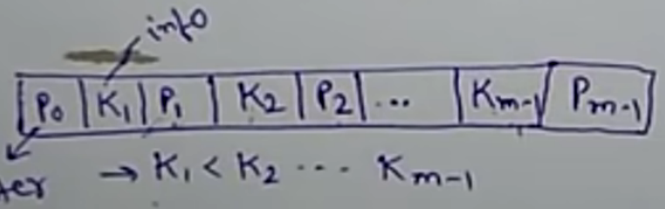

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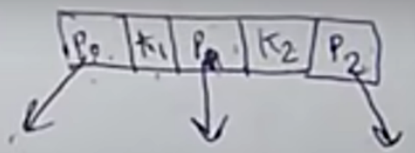

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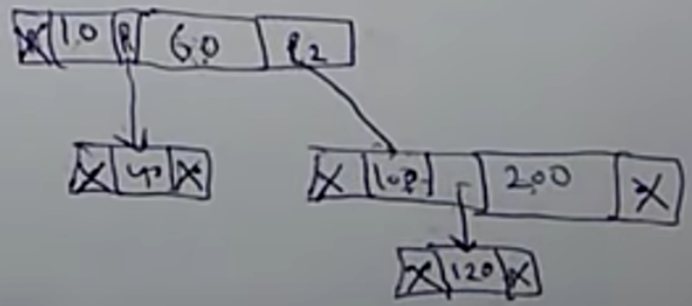

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- Generalization of BST in which a node can have more than one key & more than 2 children
- maintains sorted data
- all leaf node must be at same level
- B-tree of order m has following properties:
  - → Every node has max. m children
  - → Min. children :- leaf → 0
    root → 2
    internal nodes → $\lceil \frac{m}{2} \rceil$
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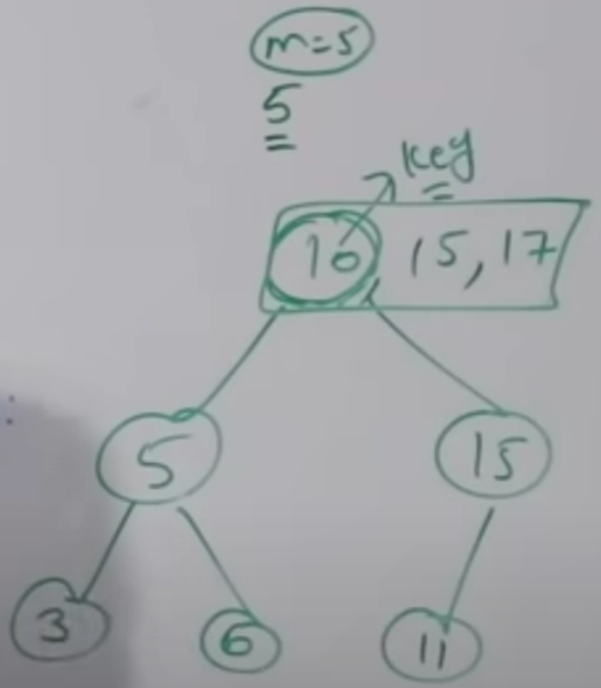

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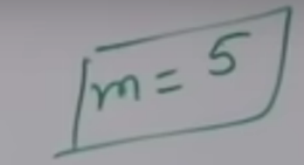

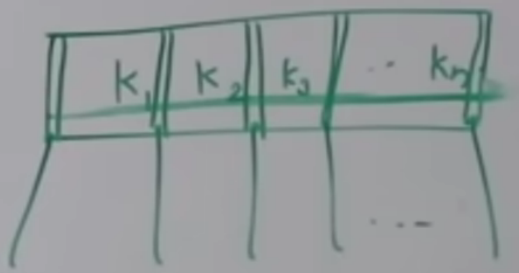

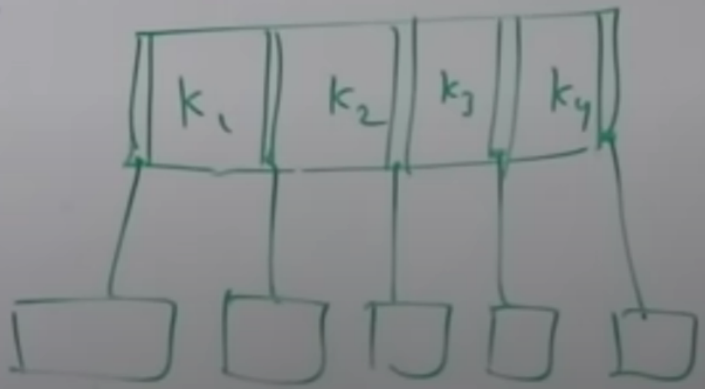

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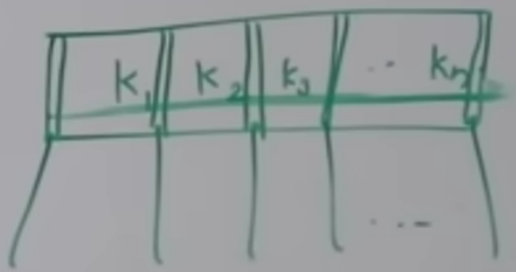
- must be at same level

m = 5



er m has following properties:

- has max. m children
- en:- leaf → 0
  root → 2
  internal nodes → $\lceil \frac{m}{2} \rceil = \lceil \frac{5}{2} \rceil = \lceil 2.5 \rceil = 3$
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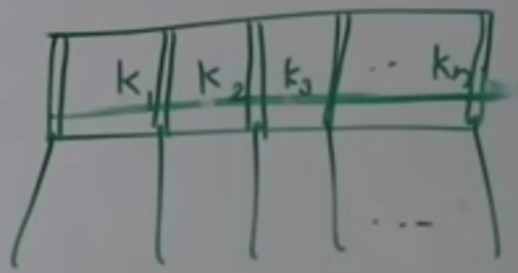

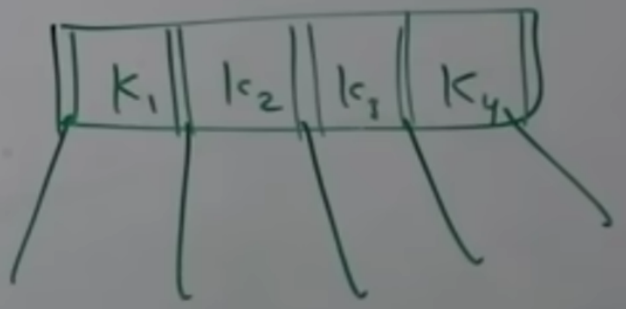

Extended Binary Tree

A binary tree T is said to be a 2-tree or extended binary tree if each node has either 0 or 2 children
A binary tree T can be converted into 2-tree by replacing each empty subtree by a new node

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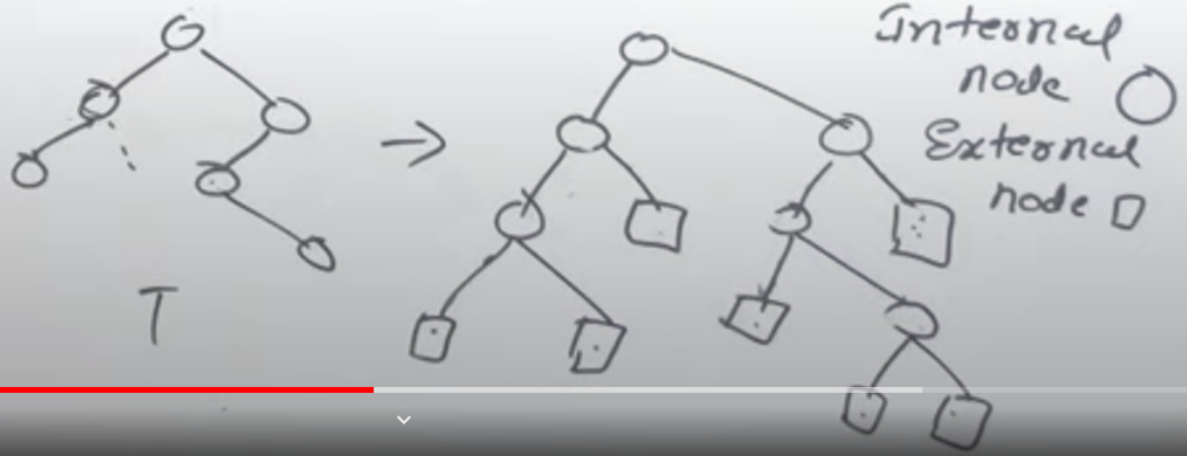
$N_E \rightarrow 7$

$N_I \rightarrow 6$

$N_E = N_I + 1$

$7 = 6 + 1$



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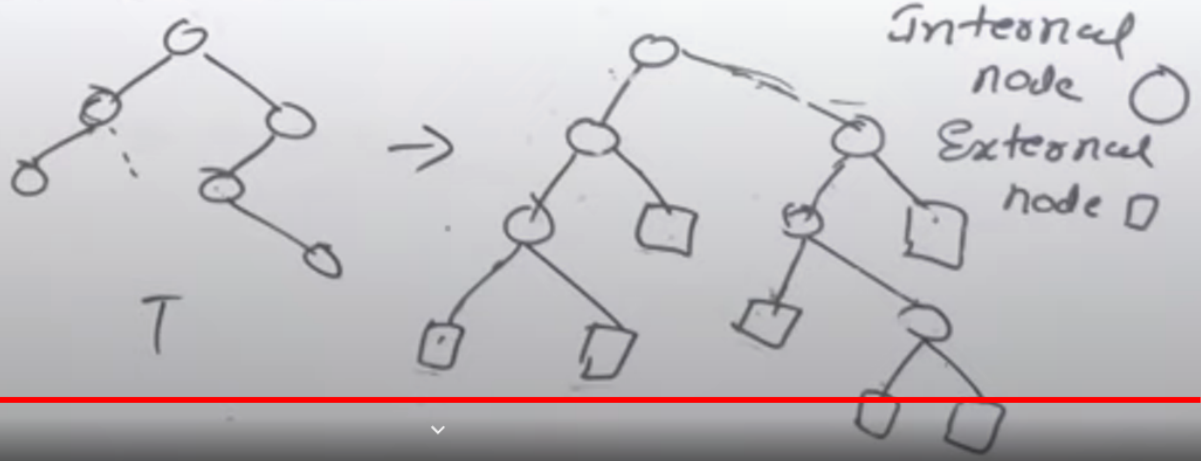
Path length

$L_I = 0+1+2+1+2+3 = 9$

$L_E = 2+3+3+3+2+4+4$

$= 21$



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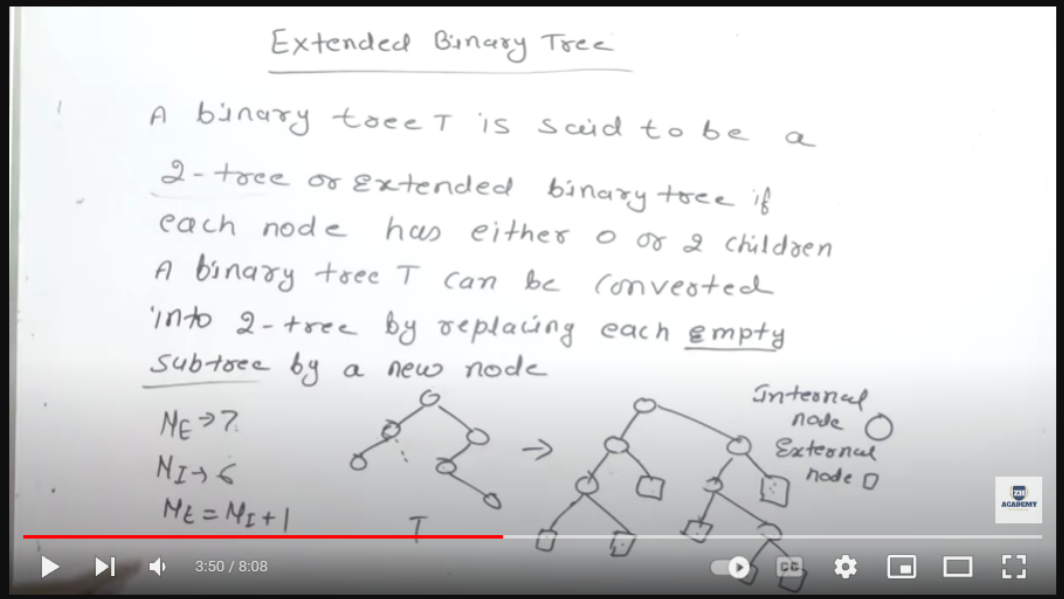

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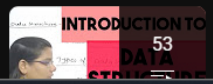

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```
185
186         where Li is path length of Internal node
187         Le is path lenght of external node.
188
189         Ni: no. of internal node.
190         Ne: no. of external node.
191
```

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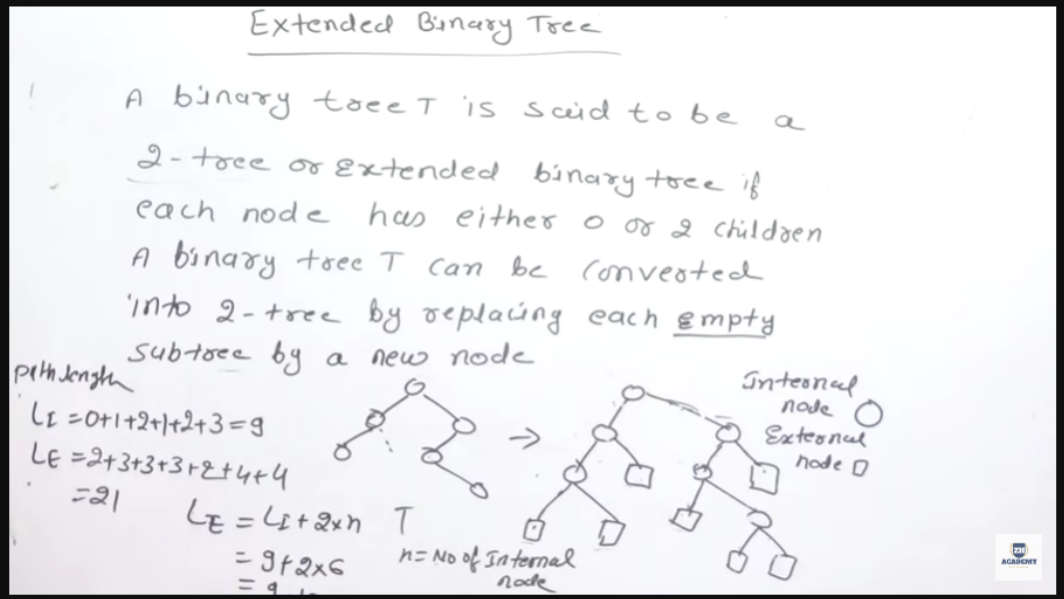

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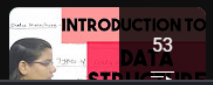

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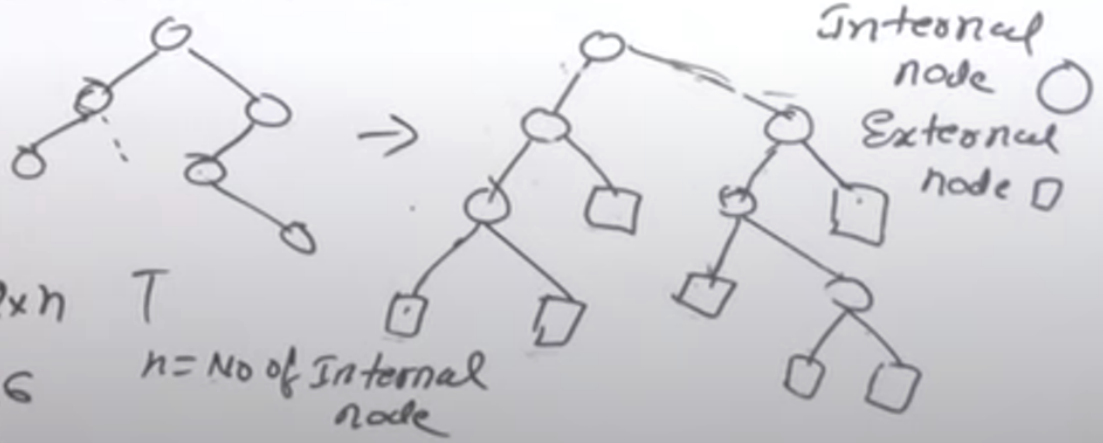

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$= 21$

$L_E = L_I + 2n$ T

$= 9 + 2 \times 6$

$= 9 + 12$

$= 21$

n = No of Internal node